TERTIARY INFOTECH ACADEMY PTE LTD
UEN: 201200696W

LESSON PLAN

For

Fundamentals of Semiconductor Device Physics and Process Integration

TGS Ref No: TGS-2024049339

Conducted by

Tertiary Infotech Academy Pte Ltd
UEN: 201200696W

Version 5.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	29 Sep 2024	First version	Tertiary Infotech Academy
2.0	23 Jul 2025	New format	Tertiary Infotech Academy
3.0	17 Oct 2025	Updated organisation name	Tertiary Infotech Academy
5.0	20 Aug 2026	Realigned to the redesigned v7.0 deck, current activity slides and 4:00-6:00 PM assessment	Tertiary Infotech Academy
5.1	20 Aug 2026	Realigned schedules and activity references to consolidated Trainer Slides v7.1	Tertiary Infotech Academy

TABLE OF CONTENTS

Course Overview

A two-day, 16-hour WSQ course comprising 14 hours of classroom facilitation and two hours of summative assessment. The learning sequence moves from device physics and fabrication architecture to process interactions, performance verification, reliability and follow-up action.

Learning Outcomes

- LO1: Identify semiconductor manufacturing process flows based on device physics.
- LO2: Determine follow-up actions for process integration issues and performance.

Day 1 Schedule - Device Physics and Fabrication

Time	Duration	Topic / Activity	Method	Slides
09:00-09:30	30 min	Digital attendance, introductions, outcomes, TSC and assessment briefing	Facilitation	1-15
09:30-10:30	60 min	Semiconductor concepts and Activity 1 briefing	Lecture / questions	16-38
10:30-10:40	10 min	Morning break	Break	-
10:40-11:15	35 min	Activity 1: Device Physics Evidence Map	Individual / peer review	423
11:15-12:10	55 min	PN junctions, diodes, MOS capacitors, MOSFETs and BJTs	Lecture / worked examples	39-104
12:10-12:50	40 min	Lunch	Break	-
12:50-14:05	75 min	Memory, wafer preparation, oxidation and RTP	Lecture / peer sharing	105-137
14:05-14:45	40 min	Activity 2: Process Module Architecture	Individual / peer review	424
14:45-14:55	10 min	Afternoon break	Break	-
14:55-16:10	75 min	Lithography, CMP, etch, doping, deposition and metallisation	Lecture / case prompts	138-248
16:10-17:00	50 min	CMOS flow and design rules	Lecture / cross-section review	249-287
17:00-17:50	50 min	Activity 3: CMOS Manufacturing Flow and Control Gates	Individual / peer review	425
17:50-18:00	10 min	Day 1 synthesis and Day 2 preview	Facilitation	17, 288

Day 2 Schedule - Integration, Verification and Action

Time	Duration	Topic / Activity	Method	Slides
09:00-09:10	10 min	Digital attendance and Day 1 retrieval	Facilitation	2, 288
09:10-09:40	30 min	Integration interactions and yield concepts	Lecture / discussion	288-329
09:40-10:25	45 min	Activity 4: Lithography-Etch Interaction Case	Case study / peer challenge	426
10:25-10:35	10 min	Morning break	Break	-
10:35-11:00	25 min	Doping and thermal evidence revisit	Lecture / worked evidence	210-228
11:00-11:45	45 min	Activity 5: Doping and Thermal-Budget Interaction	Case study / peer challenge	427
11:45-12:25	40 min	Lunch	Break	-
12:25-12:45	20 min	Performance and reliability evidence	Lecture / failure analysis	330-373
12:45-13:40	55 min	Activity 6: Yield and Performance Metric Verification	Case study / peer challenge	428
13:40-13:50	10 min	Afternoon break	Break	-
13:50-14:10	20 min	Packaging, final test and emerging integration	Lecture / examples	374-422
14:10-14:55	45 min	Activity 7: Reliability Failure Analysis	Case study / peer challenge	429
14:55-15:55	60 min	Activity 8: Follow-Up Action and Integration Review	Case study / peer challenge	430
15:55-16:00	5 min	Course close and assessment handoff	Facilitation	431-435
16:00-17:00	60 min	Written Assessment (SAQ) - open book	Assessment	-
17:00-18:00	60 min	Case Study Assessment - open book	Assessment	-

Topic-by-Topic Facilitation Map

LU1: Fundamentals of Device Physics and Device Fabrication - Slides 16-287

- Semiconductor materials and carrier physics - Slides 16-38
- PN junctions and diode behaviour - Slides 39-50
- MOSFET and BJT device operation - Slides 51-104
- Memory devices and integrated circuits - Slides 105-118
- Wafer fabrication process modules - Slides 119-248
- CMOS integration flows - Slides 249-282
- Layout and design rules - Slides 284-287

LU2: Process Integration Issues - Slides 288-422

- Isolation, wells and self-aligned modules - Slides 288-304
- Contamination, yield and process variation - Slides 305-353
- Device performance characterisation - Slides 354-362
- Device reliability and failure mechanisms - Slides 363-373
- Assembly, packaging and final test - Slides 374-406
- Optimisation and emerging technologies - Slides 407-422

Activity Facilitation Map

Activity 1: Device Physics Evidence Map - Slide 423

Purpose: Connect device-physics mechanisms to observable manufacturing controls and device consequences.

- Method: individual analysis followed by peer challenge; planned duration 35 minutes.
- Evidence: Completed device-physics evidence map with at least six mechanism-control-consequence links.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 2: Process Module Architecture - Slide 424

Purpose: Identify the major process loops, their purposes, inputs, outputs and control gates.

- Method: individual analysis followed by peer challenge; planned duration 40 minutes.
- Evidence: Process architecture map with module boundaries, control gates and two feedback loops.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 3: CMOS Manufacturing Flow and Control Gates - Slide 425

Purpose: Outline a coherent CMOS process flow and show where device physics constrains sequencing.

- Method: individual analysis followed by peer challenge; planned duration 50 minutes.

- Evidence: One-page CMOS process flow with control gates and three device-physics sequence rationales.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 4: Lithography-Etch Interaction Case - Slide 426

Purpose: Determine how variation propagates between lithography, etch, clean and contact modules.

- Method: individual analysis followed by peer challenge; planned duration 45 minutes.
- Evidence: Interaction matrix, causal chain and discriminating verification plan.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 5: Doping and Thermal-Budget Interaction - Slide 427

Purpose: Analyse implantation, activation, diffusion and oxide interactions across the thermal budget.

- Method: individual analysis followed by peer challenge; planned duration 45 minutes.
- Evidence: Thermal-budget comparison, evidence map and split-lot experiment design.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 6: Yield and Performance Metric Verification - Slide 428

Purpose: Select and interpret metrics that verify process-integration performance.

- Method: individual analysis followed by peer challenge; planned duration 55 minutes.
- Evidence: Metric dashboard worksheet with calculations, spatial interpretation and verification conclusion.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 7: Reliability Failure Analysis - Slide 429

Purpose: Verify whether TDDB, hot-carrier degradation or electromigration best explains the evidence.

- Method: individual analysis followed by peer challenge; planned duration 45 minutes.
- Evidence: Failure-mechanism comparison and reliability verification statement.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Activity 8: Follow-Up Action and Integration Review - Slide 430

Purpose: Determine proportionate containment, corrective, preventive and verification actions.

- Method: individual analysis followed by peer challenge; planned duration 60 minutes.
- Evidence: Integration review action register with containment, CAPA, owners and effectiveness checks.
- Facilitator checks the acceptance criteria and asks the learner to distinguish evidence, inference and follow-up action.

Resources Required

- Current trainer slides and learner slides
- Learner Guide
- Individual activity folders and printable PDFs
- Laptop with spreadsheet/PDF viewer
- Projector or display, whiteboard and markers
- LMS access and SSG digital-attendance QR code

Assessment

Assessment runs from 4:00 PM to 6:00 PM on Day 2: one hour for the open-book Written Assessment (K1-K2), followed by one hour for the open-book Case Study (A1-A5). Learners complete assessment digital attendance, submit answers through the LMS and sign the Assessment Summary Record.

Assessment Administration Checklist

- Confirm candidate identity, assessment digital attendance and the approved assessment version before distribution.
- Brief candidates on the one-hour duration, open-book condition, individual work requirement and C / NYC grading outcome for each instrument.
- Issue only candidate papers. Keep answer keys trainer-only and do not place answer-key links in learner-facing systems.
- Collect the Written Assessment before issuing the Case Study, then verify all pages and uploads are attributable to the candidate.
- Record K1-K2 evidence from the Written Assessment and A1-A5 evidence from the Case Study; document reassessment decisions where required.

LMS: https://lms-tms.tertiaryinfotech.com/